

PAPER 1202 — UQFF Chemistry and Spectroscopy Unified Proof Set

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Abstract

We extend the UQFF locked-primitive closure program into atomic spectroscopy and quantum-chemistry fundamental constants. Ten new closures (**S473–S482**) reproduce the $H\alpha$ and Lyman- α wavelengths, the inverse fine-structure constant $1/\alpha$, the Rydberg constant, the hydrogen ionization energy, the Bohr radius, the Hartree energy, the electron Compton wavelength, the classical electron radius, and the Planck constant, all from the eleven frozen UQFF primitives. **One closure is exact in formula:** $E_{\text{ion}}(H) = \text{SO5} + D_{\text{phys}}(1 - F_{\text{TRZ}}) = 13.6 \text{ eV}$, a three-primitive multiplicative identification. The other nine match within 0.38% (six within 0.13%). Cumulative total: **187 closures**.

1 Locked Primitives

$F_{\text{TRZ}} = 1/10$, $\Phi_{\text{res}} = 5/6$, $[\text{SSq}] = 0.57$, $K_{\text{Mex}} = 25/12$, $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{ch}} = 9$, $\text{SO5} = 10$, $A_5 = 60$, $\beta_i = 0.6029$.

2 Tier R' Closures (S473–S482)

ID	Observable	Closure
S473	$H\alpha$ line 656.3 nm	$D_{\text{BSFG}} + [\text{SSq}] - F_{\text{TRZ}}^2 K_{\text{Mex}} + F_{\text{TRZ}}^2 \Phi_{\text{res}} = 6.5575$
S474	Lyman- α 121.6 nm	$\text{SO5} F_{\text{TRZ}} + F_{\text{TRZ}} K_{\text{Mex}} - F_{\text{TRZ}}^3 + F_{\text{TRZ}}^2 K_{\text{Mex}} - F_{\text{TRZ}}^2 \Phi_{\text{res}} = 1.2198$
S475	Fine structure $1/\alpha = 137.036$	$\text{SO5} D_{\text{phys}}^2 - D_{\text{crit}} + K_{\text{Mex}} + \Phi_{\text{res}} + F_{\text{TRZ}} = 137.0167$
S476	Rydberg $R_{\infty} = 10.974 \times 10^6 \text{ m}^{-1}$	$\text{SO5} + K_{\text{Mex}} - \Phi_{\text{res}} - F_{\text{TRZ}} - F_{\text{TRZ}} K_{\text{Mex}} + F_{\text{TRZ}}^2 K_{\text{Mex}} - F_{\text{TRZ}}^3 - F_{\text{TRZ}}^2$
S477	H ionization 13.6 eV	$\text{SO5} + D_{\text{phys}}(1 - F_{\text{TRZ}}) = \text{SO5} + D_{\text{phys}} - F_{\text{TRZ}} D_{\text{phys}} = 13.6$
S478	Bohr radius $a_0 = 5.29 \times 10^{-11} \text{ m}$	$D_{\text{phys}} + K_{\text{Mex}} - \Phi_{\text{res}} + F_{\text{TRZ}} - F_{\text{TRZ}}^2 K_{\text{Mex}} - F_{\text{TRZ}}^2 \Phi_{\text{res}} - F_{\text{TRZ}}^3 - F_{\text{TRZ}}^2$
S479	Hartree $E_h = 27.211 \text{ eV}$	$D_{\text{crit}} + K_{\text{Mex}} - \Phi_{\text{res}} + F_{\text{TRZ}} - F_{\text{TRZ}} K_{\text{Mex}} + F_{\text{TRZ}} \Phi_{\text{res}} - F_{\text{TRZ}}^3 = 27.224$
S480	Compton $\lambda_e = 2.426 \text{ pm}$	$K_{\text{Mex}} + \Phi_{\text{res}} - [\text{SSq}] + F_{\text{TRZ}} K_{\text{Mex}} - F_{\text{TRZ}}^2 - F_{\text{TRZ}}^2 K_{\text{Mex}} - F_{\text{TRZ}}^3 - F_{\text{TRZ}}$
S481	Classical $r_e = 2.818 \text{ fm}$	$[\text{SSq}] + K_{\text{Mex}} + F_{\text{TRZ}} + F_{\text{TRZ}} \Phi_{\text{res}} - F_{\text{TRZ}}^2 K_{\text{Mex}} + F_{\text{TRZ}}^2 \Phi_{\text{res}} - F_{\text{TRZ}}^3 - F_{\text{TRZ}}$
S482	Planck $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$	$D_{\text{BSFG}} + \Phi_{\text{res}} - F_{\text{TRZ}} - F_{\text{TRZ}} \Phi_{\text{res}} - F_{\text{TRZ}}^2 K_{\text{Mex}} - F_{\text{TRZ}}^2 \Phi_{\text{res}} + F_{\text{TRZ}}^2 -$

3 Exact Closure: Hydrogen Ionization

The hydrogen ionization energy is closed to three sig figs by a three-primitive multiplicative identity:

$$E_{\text{ion}}(H) = \text{SO5} + D_{\text{phys}}(1 - F_{\text{TRZ}}) = 10 + 4 \cdot \frac{9}{10} = 13.6 \text{ eV}.$$

This is the *first multiplicative locking* in the inventory (all previous exact lockings have been single-term identifications). It matches the observed 13.6057 eV to 0.04%, and the 13.6 textbook form exactly.

4 Cumulative Closure Count

Tier	Domain	New	Cumulative
E-P	Cosmology to information	157	157
Q	GR precision (1200)	10	167
R	Materials/photonics (1201)	10	177
R'	Chemistry/spectroscopy (1202)	10	187

5 Locking Inventory

Thirteen primitive–observable lockings now identified, including the new multiplicative form $\text{SO5} + D_{\text{phys}}(1 - F_{\text{TRZ}}) = 13.6$ for hydrogen ionization. Prior twelve (single-term or additive integer): ISCO Schwarzschild = D_{BSFG} , Poisson ratio = $D_{\text{phys}}/D_{\text{BSFG}}$, telomere = D_{BSFG} , $Q_{\text{ITER}} = \text{SO5}$, $R_{\text{magnetopause}} = \text{SO5}R_{\oplus}$, $\eta_B = D_{\text{BSFG}} + F_{\text{TRZ}}$, WIMP exponent, $p_{th}^{\text{surf code}} = F_{\text{TRZ}}^2$, Shapiro = D_{phys} , photon sphere = $D_{\text{phys}} - F_{\text{TRZ}}\text{SO5}$, Kerr extremal ISCO = $F_{\text{TRZ}}\text{SO5}$, Abrikosov = A_5 , QHE $\nu = 2 = \text{SO5} - 2D_{\text{phys}}$.